

Requirements Clarification Knowledge Base

Reference Evaluation Dataset for Ambiguity Detection & Multi-Turn Dialogue RAG Systems

This comprehensive document serves as the master evaluation corpus and knowledge framework for building an interactive, AI-powered Business Analysis Requirements Clarification Chatbot. It contains quality frameworks, ambiguity heuristics, linguistic indicators, and detailed benchmark scenarios spanning multiple business domains (Banking, E-Commerce, Healthcare, Logistics, and Corporate IT). This multi-page text structure provides optimal material for vector chunking, metadata labeling, and embedding testing.

PART 1: CORE REQUIREMENTS QUALITY FRAMWORKS

1.1 The SMART Blueprint for Functional Requirements

Every requirement extracted or refined by the conversational agent must satisfy the SMART criteria before being deemed complete:

- **Specific:** The requirement must target a distinct system behavior or business rule without leaving room for interpretation. It explicitly identifies the active actor, the system module, and the expected boundary.
- **Measurable:** The system performance, data output, or transactional behavior must have an explicit quantitative threshold or binary pass/fail condition that a Quality Assurance (QA) engineer can test.
- **Actionable:** The technical execution must be feasible within the baseline architectural capabilities of modern computing environments without relying on impossible technical dependencies.
- **Relevant:** The stated capability must map directly to a high-level business objective, user persona goal, or compliance mandate outlined in the parent scope.
- **Time-bound / Constrained:** Performance criteria must state precise latency windowing, execution schedules, throughput bounds, or computational deadlines.

1.2 The INVEST Checklist for Agile User Stories

When requirements are processed in the form of Agile User Stories, the agent should verify structural stability using the INVEST framework:

- **Independent:** The atomic story should avoid tight functional coupling with unbuilt features.
- **Negotiable:** The requirement focuses on the business goal rather than forcing a specific hardcoded programmatic UI implementation too early.
- **Valuable:** Delivers explicit, discernible value to the end user or business enterprise.
- **Estimable:** The scope is sufficiently detailed so engineers can judge complexity and effort accurately.
- **Small:** Sized appropriately to fit within a single sprint execution cycle.
- **Testable:** Possesses unambiguous acceptance criteria that enable automation script generation.

PART 2: HEURISTICS FOR AMBIGUITY & VAGUENESS DETECTION

The NLP engine and LLM prompts must actively parse incoming stakeholder utterances for semantic red flags. The following linguistic categories denote structurally deficient statements that demand immediate multi-turn clarification questions.

2.1 Unquantified Non-Functional Performance Metrics

Vague modifiers regarding system performance fail testing standards. The agent must intercept these and mandate numeric targets:

- *"Fast" / "Quickly" / "Rapidly"*: Must be clarified to an explicit millisecond or second threshold under a defined server load (e.g., "The API response shall load within 250ms under a load of 10,000 concurrent active users").
- *"Scalable" / "High-throughput"*: Must be quantified into clear processing rates or capacity growth bounds (e.g., "The transaction pipeline must process a minimum of 1,500 messages per second during peak trading hours").
- *"Optimized" / "Minimal latency"*: Requires clarification regarding baseline network constraints and resource utilization parameters.

2.2 Subjective User Experience and Interface Terminology

Qualitative design descriptions cannot be validated programmatically. The system must ask for structural UI criteria or existing patterns:

- *"User-friendly" / "Intuitive" / "Modern Look"*: Must be translated into tangible interface patterns, maximum click paths, or accessibility compliances (e.g., "Navigation to the payment screen must take no more than 3 user clicks from the home dashboard and must fully adhere to WCAG 2.1 AA accessibility guidelines").
- *"Seamless integration"*: Must be decomposed into exact communication protocols, data format standards, or middleware specifications (e.g., "The module must push structural payload data asynchronously via an authenticated REST API over HTTPS using JSON format").

PART 3: BENCHMARK REFERENCE SCENARIOS (POOR VS. EXCELLENT MAPPINGS)

Use these concrete operational scenarios to populate vector database embeddings. This allows the RAG agent to dynamically draw semantic parallels between raw user input and verified logic gaps.

3.1 Domain A: Core Banking & Wealth Management

SCENARIO 1: FRAUD DETECTION LOGIC

Deficient Input: "The system needs to look out for suspicious account transfers and freeze things automatically to keep money safe."

Logic & Ambiguity Gaps: Identifies no clear trigger for what makes a transfer "suspicious," fails to specify what "freeze things" means (Does it block the login, the specific card, or the transaction pipeline?), and uses highly ambiguous phrasing ("automatically to keep money safe").

Refined Requirement: "The Fraud Core Engine shall flag any single outbound domestic transfer exceeding \$10,000 USD or any international transfer originating from an IP address outside the user's home country profile as high-risk. Upon flagging, the system must hold the execution state for up to 30 minutes, transition the transaction status to 'PENDING_VERIFICATION', and instantly push a high-priority SMS and email push notification containing a 6-digit One-Time Passcode (OTP) to the account holder."

SCENARIO 2: END-OF-DAY FINANCIAL REPORTING

Deficient Input: "Managers should get a summary report of daily ledger balances sent to them as soon as possible after trading closes."

Logic & Ambiguity Gaps: The actor "Managers" is a loose, generic term lacking role-based access control. "As soon as possible" is unquantified. The transport channel and fallback mechanisms are omitted entirely.

Refined Requirement: "The Ledger Reconciliation Module shall generate a daily balancing report in CSV and PDF formats at exactly 17:15 EST, fifteen minutes after the New York stock exchange trading floor closure. The system must distribute this report via encrypted email to the 'Risk_Operations_Manager' role group. If the ledger balances show an unresolved variance exceeding \$0.01 USD, an automated critical alert must be routed to the Chief Financial Officer's dashboard."

3.2 Domain B: E-Commerce & Retail Supply Chain

SCENARIO 3: MULTI-TIERED DISCOUNT PRICING ENGINE

Deficient Input: "Give good discounts to our frequent shoppers when they buy a bunch of clothing items together at checkout, etc."

Logic & Ambiguity Gaps: Relies on an unbounded list modifier ("etc."). Uses highly ambiguous qualifiers ("good discounts", "frequent shoppers", "bunch of clothing items"). Provides zero mathematical formulas for calculation.

Refined Requirement: "The Pricing Engine shall apply a 15% line-item discount on all products categorised under 'Apparel' during checkout processing, provided that the authenticated customer holds a 'Platinum Loyalty' tier status and the cart contains a minimum of five distinct apparel SKU units. This discount calculation must execute in memory before tax calculation and must be logged as a separate field in the checkout payload structure."

3.3 Domain C: Healthcare Information Systems (HIS)

SCENARIO 4: ELECTRONIC HEALTH RECORD (EHR) PATIENT CHART ACCESS

Deficient Input: "The app should show patient health records super fast, but make sure it is totally locked down so bad guys can't see the health data."

Logic & Ambiguity Gaps: Uses colloquial speed descriptors ("super fast") and generic security concepts ("totally locked down", "bad guys"). Fails to declare compliance mandates like HIPAA or technical identity frameworks.

Refined Requirement: "The EHR Viewer Module shall render the historical patient medical chart data in under 1.5 seconds from initial query execution. All data transmission must be encrypted in transit using TLS 1.3. User authentication must be verified via Active Directory, and access token validation must enforce role-based permissions, restricting data views strictly to 'Licensed_Medical_Practitioners' assigned to that specific patient's active care team, in full compliance with HIPAA Privacy Rules."

PART 4: TARGET DATA STRUCTURE FOR EMBEDDINGS (JSON ENCODING)

To maximize RAG matching effectiveness, index the metadata alongside the raw texts. The chatbot can then leverage these keys during vector similarity matching:

```
{
  "scenario_id": "REQ-REF-042",
  "domain": "Capital Markets",
  "ambiguity_tags": ["quantitative_latency", "role_definition_gap", "unbounded_list"],
  "target_heuristics": ["As soon as possible", "Managers", "etc"],
  "clarification_rule": "Intercept the phrase 'as soon as possible' and request a
distinct timeframe in minutes or seconds. Force the stakeholder to explicitly map
'Managers' to a strict role group within the system access registry.",
  "synthetic_variation": "Our staff wants the balance sheet updates immediately when
trading ends, plus other sheets etc."
}
```